Biometric Matching

Biometric Matching Process

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Step 1: Capture biometric sample from sensor

Step 2: Pre-process the sample (enhancement, normalization)

Step 3: Extract features and create query template

Step 4: Retrieve stored template(s) from database

Step 5: Apply matching algorithm to compute similarity score

Step 6: Compare score against threshold and make decision

Classification of Matching Methods

Correlation-based

- Approach
  - Direct comparison of biometric images or signals
- Method
  - Pixel to pixel or signal correlation
- Examples
  - Cross-correlation
  - Normalized correlation
  - Phase correlation
- Best for
  - Raw image matching, retina patterns

Feature-based

- Approach
  - Extract and compare distinctive features
- Method
  - Feature vector comparison
- Examples
  - Minutiae matching(Fingerprint)
  - Landmark matching(face)
  - Texture pattern matching(Iris)
- Best for
  - Most biometric modalities

Key Difference: Correlation methods work with raw data (computationally expensive), while feature-based methods work with extracted features (more efficient and robust).

Distance Metrics for Matching

- Distance metrics measure dissimilarity between templates

  Lower distance = Higher similarity.

1. Euclidean Distance

- **Formula:** $d(P, Q) = \sqrt{\Sigma(p_i - q_i)^2}$
- Most common distance metric
- Measures straight-line distance in feature space

2. Manhattan Distance (L1 Norm)

- **Formula:** $d(P, Q) = \Sigma|p_i - q_i|$
- Sum of absolute differences
- Less sensitive to outliers than Euclidean

3. Hamming Distance

- **Formula:** $d(P, Q)$ = Number of positions where bits differ
- Used for binary feature vectors (e.g., iris codes)
- Very fast computation

Advanced Distance Metrics

4. Mahalanobis Distance

- **Formula:** $d(P, Q) = \sqrt{((P-Q)^T \Sigma^{-1} (P-Q))}$
- Accounts for correlations between features and different scales
- Σ is covariance matrix

5. Cosine Similarity

- **Formula:** $\text{similarity} = (P \cdot Q) / (\|P\| \times \|Q\|)$
- Measures angle between vectors
- Value ranges from -1 to 1 (1 = identical direction)

6. Chi-Square Distance

- **Formula:** $d(P, Q) = \Sigma((p_i - q_i)^2 / (p_i + q_i))$
- Used for histogram-based features (e.g., texture descriptors)

Fingerprint Matching methods

- Advantages
  - Small template size
  - Widely standardized
  - Works with partial prints
  - Rotation/Translation invariant
- Challenges
  - Sensitive to image quality
  - Minutiae detection errors
  - Alignment complexity
  - Performance degrades with dry/wet fin

1. Minutiae-Based Matching

- **Features:** Ridge endings and bifurcations
- **Template:** Set of (x, y, θ) coordinates for each minutia point

Matching Algorithm:

1. Align templates (rotation and translation)
2. Find corresponding minutiae pairs
3. Count matched minutiae within tolerance
4. Compute match score = (matched pairs) / (total minutiae)

Fingerprint Pattern-Based Matching

2. Correlation-Based Matching Direct comparison of fingerprint images using correlation.

Formula: Correlation = $[\Sigma \Sigma I_1(x,y) \times I_2(x,y)] / \sqrt{(\Sigma \Sigma I_1^2 \times \Sigma \Sigma I_2^2)}$

3. Ridge Feature-Based

- **Ridge Orientation:** Direction of ridges at each point
- **Ridge Frequency:** Number of ridges per unit distance
- **Ridge Shape:** Curvature and flow patterns

4. Hybrid Methods Combine minutiae + texture features for improved accuracy

Comparison Table:

Method	Accuracy	Speed	Storage
Minutiae-based	High	Fast	Small
Correlation-based	Very High	Slow	Large
Hybrid	Very High	Medium	Medium

Face Recognition Matching Methods

1. Geometric/Feature-Based Extract facial landmarks (eyes, nose, mouth) and compute geometric relationships

Features:

- Distance between eyes
- Nose width and length
- Face width and height
- Jaw line shape

Matching: Euclidean distance on feature vector

2. Appearance-Based (Holistic) Treat entire face as a pattern

Two Sub-methods:

Eigenfaces (PCA)

- Project faces onto principal components
- **Match:** Euclidean distance in eigenspace

Fisherfaces (LDA)

- Maximize class separability
- **Match:** Mahalanobis distance

Modern Face-recognition –Deep Learning

3. Deep Learning –based matching

Process(4 Steps)

Step1: Face Detection- Locate face in image

Step2: Face Alignment- Normalize pose and scale

Step3: CNN Feature Extraction -128-512 dimensional embedding

Step4: Matching- Cosine similarity or Eucliden distance

Popular Architecture

- FaceNet: Triplet loss learning for embeddings
- DeepFace: 9-layer deep network by facebook
- VGGFace: Based on VGG-16 architecture
- ArcFace: Additive angular margin loss

Advantage: Achieves >99% accuracy on standard benchmarks

Iris Recognition Matching

Daugman's Algorithm (Most Popular)

Process (4 Steps):

Step 1: Iris Segmentation - Detect inner and outer boundaries

Step 2: Normalization - Convert to rectangular format (polar coordinates)

Step 3: Feature Encoding - Apply 2D Gabor wavelets

Step 4: Create IrisCode - 2048-bit binary template

Matching Process:

Formula: Hamming Distance = (XOR of two iris codes) / (Total bits)

Decision: If HD < 0.32, declare match (typically)

Why Hamming Distance? Binary iris codes allow extremely fast matching (millions of comparisons per second). XOR operation is hardware-optimized.

Voice Recognition Matching

Feature Extraction:

- **MFCC:** Mel-Frequency Cepstral Coefficients (most common)
- **LPC:** Linear Predictive Coding
- **Pitch and Formants:** Fundamental frequency characteristics

Matching Approaches:

1. Template Matching

DTW (Dynamic Time Warping):

- Aligns sequences of different lengths
- Finds optimal path through distance matrix
- Good for text-dependent systems

2. Statistical Models

GMM (Gaussian Mixture Model):

- Model voice as mixture of Gaussians
- Compute likelihood of test sample
- Text-independent recognition

3. Deep Learning (i-vectors & x-vectors) Extract fixed-dimensional embeddings, then use cosine similarity for matching

Signature Verification Matching

Static (Offline)

- Image of signature on paper
- Features:
  - Shape of descriptors
  - Geometric features
  - Pressure patterns
  - Texture Anlysis

Matching: Correlation or feature distance

Dynamic (Online)

- Captured during process
- Features:
  - Pen position (x, y)
  - Velocity and acceleration
  - Pressure and angle
  - Timing information

Matching: DTW on temporal sequences

Note: Dynamic signatures are much more secure (harder to forge) but require special hardware.

Comparative Analysis of Matching methods

Method	Accuracy	Speed	Memory	Best For
Correlation	High	Slow	High	Small databases, high security
Minutiae (FP)	High	Fast	Low	Fingerprint systems
Hamming (Iris)	Very High	Very Fast	Very Low	Large-scale iris recognition
Eigenfaces	Medium	Fast	Medium	Controlled face recognition
Deep Learning	Very High	Fast*	High	Modern face, fingerprint systems
DTW (Voice)	Medium-High	Medium	Medium	Voice verification

- After feature extraction; GPU required for real-time performance

Best practices for Biometric Matching

Step 1: Always normalize scores before fusion or comparison

Step 2: Implement quality checks - reject poor quality samples early

Step 3: Use multiple enrollment samples for robust templates

Step 4: Set thresholds based on application requirements, not defaults

Step 5: Log match scores and decisions for performance monitoring

Step 6: Test with diverse population representing actual users

Step 7: Implement liveness detection to prevent spoofing

Step 8: Provide clear feedback to users for failed matches